THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS FULL SYLLABUS PAPER SET 05

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The point $(3,\ 0)$ is reflected in the x-axis. Its image is: 1

(a) $(0,\ 3)$ **(b)** $(-3,\ 0)$ **(c)** $(3,\ 0)$ **(d)** $(0,\ -3)$

2. The point $(8,\ 1)$ lies: 1

(a) Quadrant I **(b)** Quadrant II **(c)** Quadrant IV **(d)** Quadrant III

3. The distance between $A(-2,\ 3)$ and $B(3,\ -9)$ is: 1

(a) 12 units **(b)** 15 units **(c)** 14 units **(d)** 13 units

4. The value of $7x^2 - 4x + 6$ at $x = 0$ is: 1

(a) 11 **(b)** 3 **(c)** 8 **(d)** 6

5. The sequence $3, 7, 11, 15, \ldots$ forms a linear pattern. Its nth term is: 1

(a) $4n - 1$ **(b)** $4n + 3$ **(c)** $3n + 4$ **(d)** $4n - 7$

6. The decimal expansion of $\frac{4}{15}$ is: 1

(a) terminating **(b)** non-terminating repeating

(c) non-terminating non-repeating **(d)** cannot be determined

7. The number $\frac{22}{7}$ is: 1

(a) not a real number **(b)** rational

(c) neither rational nor irrational **(d)** irrational

8. The number $0.\overline{6}$ in the form $\frac{p}{q}$ is: 1

(a) $\frac{2}{3}$ **(b)** 1 **(c)** $\frac{3}{2}$ **(d)** $\frac{23}{30}$

9. Factorise $x^2 - 5x - 36$: 1

(a) $(x - 9)(x - 4)$ **(b)** $(x - 4)(x + 9)$ **(c)** $(x - 9)(x + 4)$ **(d)** $(x + 4)(x + 9)$

10. The expression $36x^2 + 12x + 1$ can be factorised as: 1

(a) $(6x + 2)^2$ **(b)** $(6x + 1)^2$ **(c)** $(6x - 1)^2$ **(d)** $(36x + 1)^2$

11. The sum of either pair of opposite angles of a cyclic quadrilateral is: 1

(a) 270° **(b)** 180° **(c)** 360° **(d)** 90°

12. ABCD is a cyclic quadrilateral with $\angle A = 75^\circ$. Then $\angle C$ is: 1

(a) 53° **(b)** 105° **(c)** 75° **(d)** 285°

13. A chord of length 16 cm is drawn in a circle of radius 17 cm. The distance of the chord from the centre is: 1

(a) 17 cm **(b)** 15 cm **(c)** 9 cm **(d)** 25 cm

14. The length of an arc subtending 30° at the centre of a circle of radius 42 cm ($\pi = \frac{22}{7}$) is: 1

(a) 264 cm **(b)** 44 cm **(c)** 22 cm **(d)** 11 cm

15. In Heron's formula, s stands for: 1

(a) the area of the triangle **(b)** the semi-perimeter of the triangle
(c) the longest side **(d)** the perimeter

16. The perimeter of a circle is 132 cm. Taking $\pi = \frac{22}{7}$, its radius is: 1

(a) 42 cm **(b)** 21 cm **(c)** 22 cm **(d)** 10.5 cm

17. A fair die is rolled once. The probability of getting a factor of 12 is: 1

(a) $\frac{5}{7}$ **(b)** $\frac{5}{6}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{2}{3}$

18. A bag contains 9 red, 3 blue and 3 green balls. One ball is drawn at random. The probability that it is red is: 1

(a) $\frac{3}{5}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{4}{5}$ **(d)** $\frac{2}{5}$

19. The sequence $2, 5, 8, 11, \ldots$ is: 1

(a) a GP with $r = 3$ **(b)** an AP with $d = 3$
(c) neither an AP nor a GP **(d)** an AP with $d = 2$

20. The 5th term of the GP with first term 2 and common ratio 3 is: 1

(a) 162 **(b)** 163 **(c)** 161 **(d)** 324

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 20–30 words each.

21. In which quadrant (or on which axis) do the points $(-2,\ 5)$, $(6,\ -3)$, $(0,\ -4)$ and $(-1,\ -1)$ lie? 2

22. State the slope and y-intercept of $y = -4x + 9$ and say whether it shows linear growth or decay. 2

23. A rational number in lowest form has denominator $2^3 \times 5$. How many decimal places does its expansion have? Why? 2

24. Evaluate 23×17 using a suitable identity. 2

25. Points C and D lie on the same side of AB and $\angle ACB = \angle ADB$. What can you say about A, B, C, D? 2

26. Among all rectangles with perimeter 40 units, which has the greatest area? State that area. 2

27. If the probability that a team wins is 0.35, what is the probability that it does not win? Why? 2

28. Find the sum of the first 20 natural numbers and of the first 100 natural numbers. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. The points $(1,\ -8)$, $(-4,\ 7)$ and $(-7,\ -4)$ lie on a circle with centre at the origin. Find its radius and say whether $(-5,\ 6)$ lies inside, on or outside the circle. 3

30. The area of a rectangle of length 13 cm and breadth b cm is a linear polynomial in b. Find it for $b = 12, 10, 8$ and write the pattern. 3

31. Classify as rational or irrational: 3
(i) $\sqrt{81}$
(ii) $\sqrt{12}$
(iii) $0.\overline{3}$
(iv) $0.12345\,12345\ldots$ (repeating block). Give the fraction where rational.

32. Is $(a+b-c)^2 + (a-b+c)^2 + (a-b-c)^2 = 2a^2 + 2b^2 + 2c^2$ an identity? Test with $a = 1, b = 1, c = 1$. 3

33. Baudhāyana's method: find the side of a square whose area equals that of a $16\ \text{cm} \times 9\ \text{cm}$ rectangle. 3

34. A card is drawn from a pack of 52. Find the probability of 3
(i) a black queen
(ii) the ace of spades
(iii) a number card from 2 to 10.

35. In cyclic quadrilateral ABCD, $\angle A = (4x - 4)^\circ$ and $\angle C = (2x + 16)^\circ$. Find x and both angles. 3

OR

Explain why three collinear points cannot lie on a circle. 3

36. Show that the points $(1, 3), (2, 6), (3, 12)$ from the GP $3, 6, 12, \ldots$ do not lie on a straight line. 3

OR

Which term of the GP $2, 6, 18, \ldots$ is 486? 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. Find 5 rational numbers between $\frac{3}{5}$ and $\frac{4}{5}$ by writing them with a common denominator large enough. Explain why the common denominator must be at least 6 times the difference of the numerators. 5

OR

(i) Prove that $\sqrt{3}$ is irrational. 5
(ii) Hence prove that $2 - \sqrt{3}$ is irrational.

38. Prove that the perpendicular bisector of a chord of a circle passes through the centre. 5

OR

Case study: Points A(0,0), B(8,0), C(0,6) and D(8,6) are corners of a rectangular plot. 5
(i) Find the length of diagonal BC.
(ii) Find the centre and radius of the circle through A, B, C.
(iii) Show that D lies on the same circle.
(iv) What type of quadrilateral is ABDC and why is it cyclic?

39. Prove that the area of a triangle is $\frac{1}{2}bh$ and deduce that a median divides it into two triangles of equal area. 5

OR

(i) Using Heron's formula show that the area of an isosceles triangle with equal sides a and base $2b$ is $b\sqrt{a^2 - b^2}$. 5
(ii) Check with $a = 5, b = 3$ by the formula $\frac{1}{2}bh$.

40. **Case study:** A taxi company charges a fixed booking fee of ₹200 plus ₹40 per km. 5

(i) Write the fares for 1, 2 and 3 km.

(ii) Show the sequence is an AP and find the nth term.

(iii) Find the fare for 10 km.

(iv) For how many km is the fare ₹1,000?

OR

Case study (Sierpiński triangle): Stage 0 is a black triangle of area 1 sq. unit. 5

(i) Give the number of black triangles at Stages 0 to 3.

(ii) Give the number at Stages 4 and 5.

(iii) Write the rule for Stage n.

(iv) Find the black area at Stage 3 and at Stage n.

(v) What happens to the area as n grows?

END OF QUESTION PAPER **Total: 80 Marks**

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